Calibration

- Requirements
- Process control terminologies*
- Calibration of instruments
  - Instrument errors
  - Calibration errors
  - Instrument signals

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Learning Outcomes

To achieve the following Learning Outcomes:

- Have knowledge and understanding of the various process industry instruments, the concept of measurements, calibration and configuration requirements and their applications.
- Be able to design and develop a control loop consisting the process instruments, based on a prescribed requirement.

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Assessment Criteria

Assessment criteria:

To demonstrate student has achieved the learning outcomes

- Describe the measuring principles, calibration and configuration requirements of different process industry instruments and their applications.
- Design and develop a control loop using the relevant instruments to meet specifications

Process , Industrial Instrumentation & Measurement

Assessment criteria:

- **Basic concepts of measurements**
- **Process control terminologies**
- **Calibration**

Assessment criteria:

- Applications and selections of sensors and actuators used in industries
- Pressure Level Temperature Flow

Lab experiences

- Calibration, configuration of transmitters, controllers, recorders and other peripherals, wiring, selection and installation.
- Design and implementation of a simple control loop.

Assessment criteria:

- Signal Conditioning
- P,T,L,F to mA, V
- V-V,V-mA,mA-V

Assessment criteria:

- Design and implement a simple control loop.

DESIGN CONSIDERATIONS

INSTRUMENT ACCURACY REQUIREMENT

1. **TURBINE METER- +/-0.25 LINEARITY, +/-0.02% REPEATABILITY**
2. **PRESSURE TRANSMITTER- +/- 0.25% SPAN**
3. **TEMPERATURE TRANSMITTER- +/- 0.25% OF SPAN**
4. **BALANCER (MASS)- ZERO DRIFT 0.25mm/^{o}C, SENSITIVITY DRIFT 1.0 mm/^{o}C**

Why Calibration is required?

All Instrumentation drifts or fails after a period of time, some instruments fail sooner rather than later. In order to keep measurement accuracy instruments need regular calibration against known standards.

Obviously you can't carry national standards around so you calibrate against an instrument traceable to these national standards so all the instruments must be calibrated against instrument traceable to national standards

<u>OBJECTIVE</u>

The objective is to verify the measurement accuracy of any particular instrument that contributed to the quantification of medium (liquid/gas etc) F/L/T/P at operating conditions is within the legally required limits

CALIBRATION AND VALIDATION

- **System Validation**

  - Conduct periodic calibration and validation at agreed frequency. To maintain its required accuracy and integrity.
  - Establish an approved validation manual and agreed by all relevant parties before start-up
  - Validation report to be prepared.

Process control terminology

1	**Accuracy**: is the degree of conformity of an indicated value to a recognized accepted standard value
2	Accuracy rating: is a number that defines a limit that errors will not exceed when a device is used under specified operating conditions
3	Ambient Temperature: is the temperature of the medium surrounding a device
4	Angularity: same as linearity
5	**Calibrate**: is to ascertain output of a device corresponding to a series of values of the quantity the device is to measure, receive, or transmit
6	Controlled variable: is the process variable being controlled

Process control terminology

7	Deadband: is the range through which an input can be varied without initiating an observable response. Deadband is usually expressed in percentage of span.
8	Deviation: any departure from a desired or expected value or pattern
9	**Drift**: an undesirable change in the output-input relating over a period of time.
10	Elevated zero range: is a range where the zero value of the range is greater than the lower range value, eg., -25 to 100, -200 to –20, -100 to 0
11	Error: is the algebraic difference between the indication and the ideal value of the measured signal. Error = Indication minus Ideal value.
12	**Hysteresis**: is the maximum difference for the same input between the upscale and down scale output values during a full range travel in each direction.

Process control terminology

13	**Linearity**: is the closeness to which a curve approximates a straight line
14	Lower and upper value (LRV, URV): is the lowest and highest value of the measured variable that the device can be adjusted to measure
15	Manipulated variable: is the variable which is adjusted to maintain a constant value of the controlled variable
16	Measured variable: is the physical quantity, property, or condition which is to be measured
17	Measured signal: is a signal produced by the primary element and applied to the input of secondary element
18	Output signal: is a signal delivered by a device, element or system

Process control terminology

19	**Precision**: is an instrument's degree of freedom from random errors. If a large number of readings are taken of the same quantity by a high-precision instrument, then the spread of readings will be very small. A high precision instrument may have a low accuracy.
20	Process: any operation or sequence of operations involved in converting raw material into the required product
21	**Range**: is the region between limits within which a quantity is measured. Expressed by stating the lower and upper range values. Eg., 0% to 100%, 25°C to 75°C, 4mA to 20 mA)
22	**Repeatability**: is the closeness of agreement among a number of consecutive calibration checks,at the same values under the same conditions, approaching from the same direction.
23	Resolution: is the least interval between the adjacent discrete details which can be distinguished one from the other . OR is the degree of precision (clarity) of an indicated values.

Process control terminology

24	**Span**: is the difference between the upper and lower range values Eg., range 0 to 100%, SPAN is 100 range 25 to 75 ^{o} C, SPAN is 50 range 4 to 20mA, SPAN is 16
25	Span error: is the error in calibration at the upper range value
26	**Sensitivity**: is the ratio of a change in output magnitude to the change in input (after steady state)
27	Suppressed zero range: is a range where the zero value of the range is less than the lower range value. Eg., 4 to 20, 3 to 15, 20 to 100.
28	Threshold: is the smallest change in the input signal that will result in a measurable change in the output signal

Process control terminology

29	Transducer: is a device that receives information from one system and generates an output in response to it
30	Transmitter: is a transducer that responds to a measurement variable and converts that input into a standardized transmission signal
31	**Zero error**: is the error in calibration at the lower range value
32	Zero based range: is the range that has zero as its lower range value Eg., 0 to 1000, 0 to 125, 0 to 7.5

How to perform Calibration?

'Calibration procedure' involves using a standard sample in place of the measured quantity as the input to the instrument, and obtaining the relationship between the output (reading) and the true value.

Standard : the known value used for calibration

- *Objective of calibration*

To establish the relationship between the value of the input to the measurement system and the system's indicated output value: zero input values matches the zero output value, max input value matches max output value, and mid-point input value matches with mid-point output value.

These values as well as errors such as 'repeatability' and 'hysteresis' will be within the limits (or accuracy) specified by the manufacturer.

Instruments errors

Range:

The range of an instrument is usually regarded as the difference between the maximum and minimum reading. For example a thermometer that has a scale from 20 to 100 ^{o}C has a range of 80 ^{o}C. This is also called the full scale deflection (f.s.d).

- **Operating Range (input span, output span)**
  - The region within which a quantity is measured. Expressed by stating the lower and upper range values.

    eg. 4mA to 20 mA, 25°C to 75°C, 10% to 90%.
  - The input operating range, *input span* is defined as extending x_{min} to x_{max}

$$r_i = x_{\max} - x_{\min}$$

  - The output operating range, *output span(full-scale operating range)* is specified from y_{min} to y_{max} and expressed as

$$r_o = y_{\max} - y_{\min}$$

Instruments errors

Accuracy:

The accuracy of an instrument is often stated as a % of the range or full scale deflection. For example a pressure gauge with a range 0 to 500 kPa and an accuracy of plus or minus 2 % f.s.d. could have an error of plus or minus 10 kPa. When the gauge is indicating 10 kPa the correct reading could be anywhere between 0 and 20 kPa and the actual error in the reading could be 100 %. When the gauge indicates 500 kPa the error could be 2% of the indicated reading.

- The accuracy of the measurement system is refers to its ability to indicate a true value exactly.
- Accuracy is related to *absolute error,* $\in$, defined as the difference between the true value applied to measurement system and the indicated value of the system.

$$\in = true\ \ value - indicated\ \ value$$

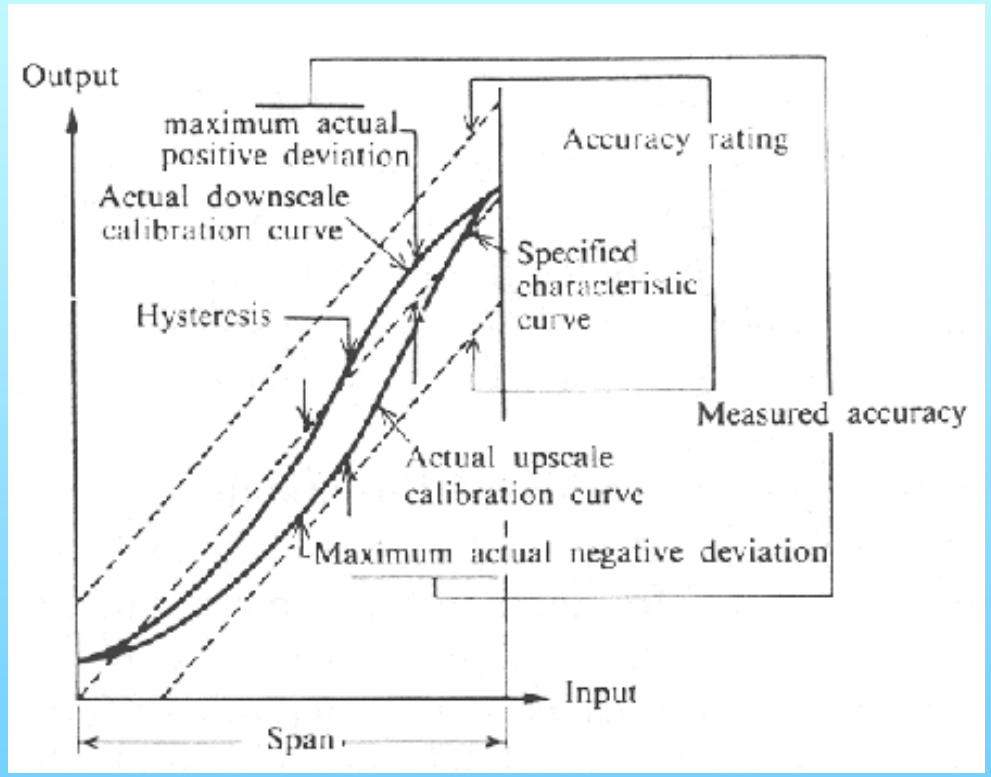
Instruments errors

- *% accuracy, A*

$$A = \left(1 - \frac{\in}{true \quad value}\right) \times 100$$

- An alternative form of calibration curve is the deviation plot.
- Deviation curves are extremely useful when the differences between the true and the indicated value are too small to suggest possible trends on direct calibration plots.

Instruments errors



Accuracy

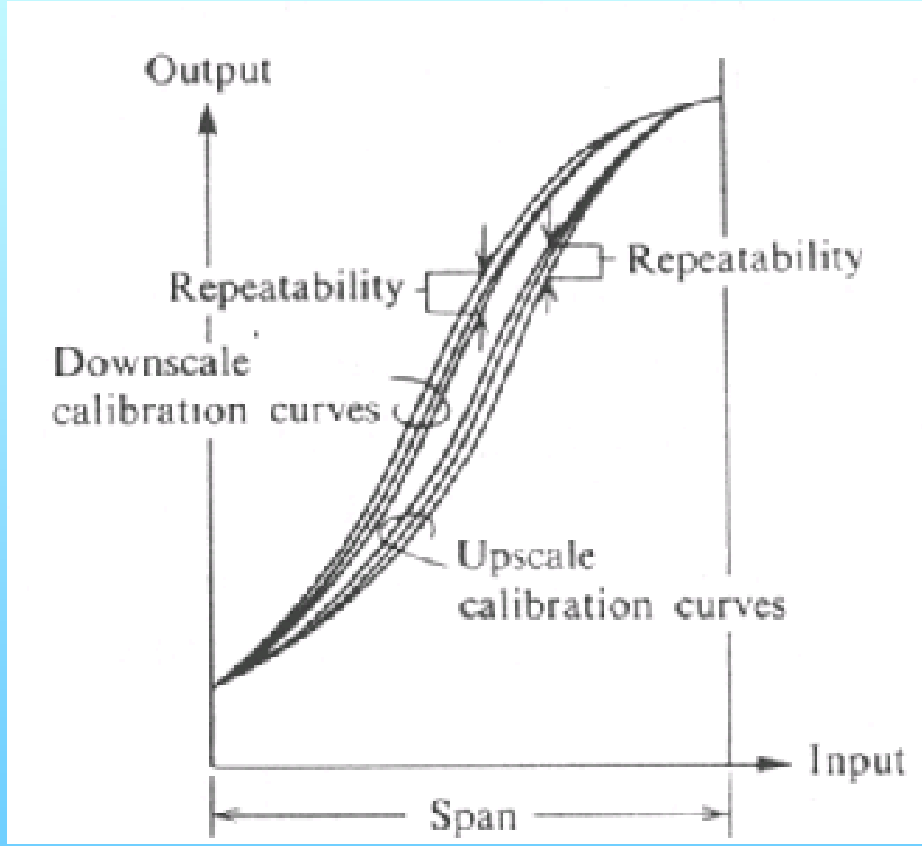
Accuracy includes:

- Effect of hysteresis (difference in output according to direction of change in the input).
- Dead zone (the largest change in input that fails to produce any output).
- Repeatability.

Instruments errors

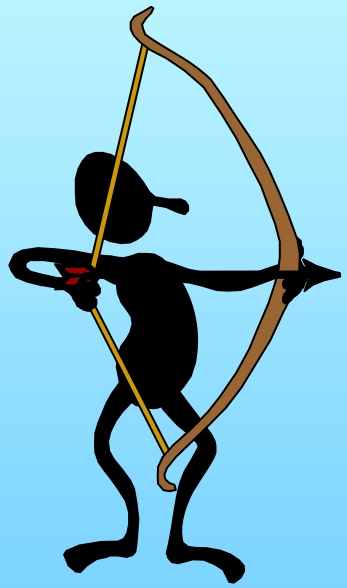
Repeatability

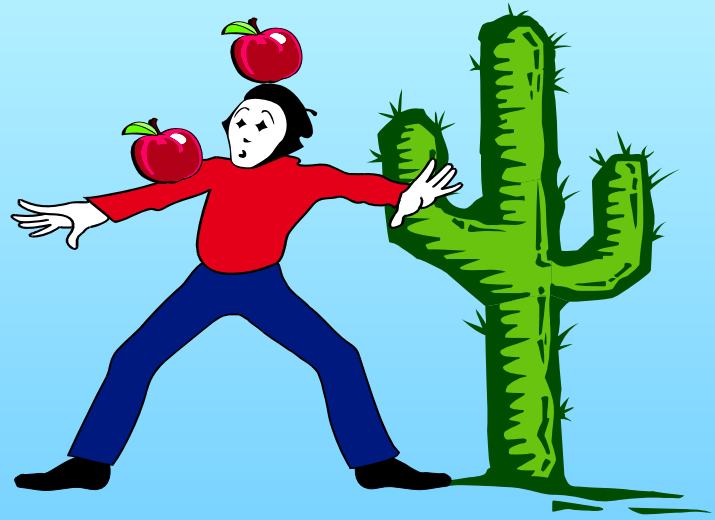
The degree to which measurements of the same object made by the same method, under the same conditions, and repeated within a relatively short period of time, produce the same measured values, all of which are causes of error, see graph.



Repeatability

Illustrations on repeatability, accuracy, precision …

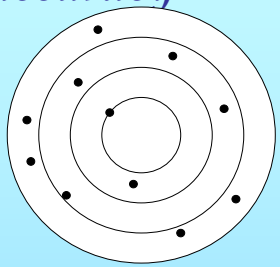




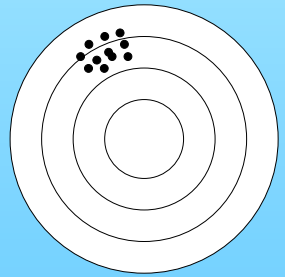
Repeatability , Accuracy, Precision

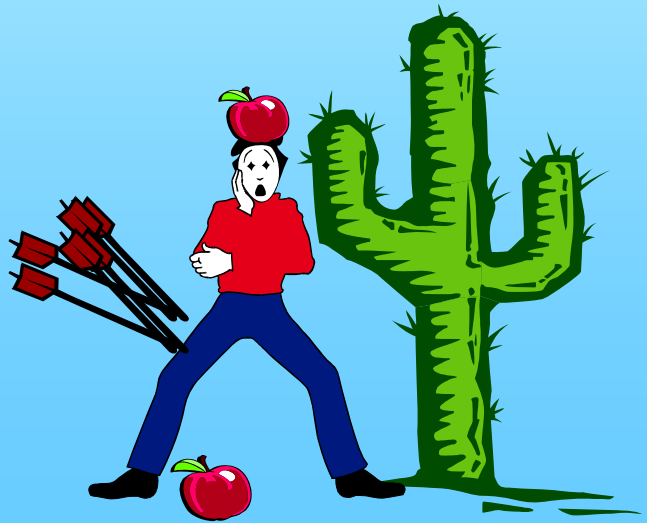
Poor repeatability
means poor accuracy
Low precision, low accuracy

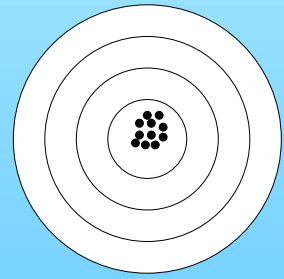


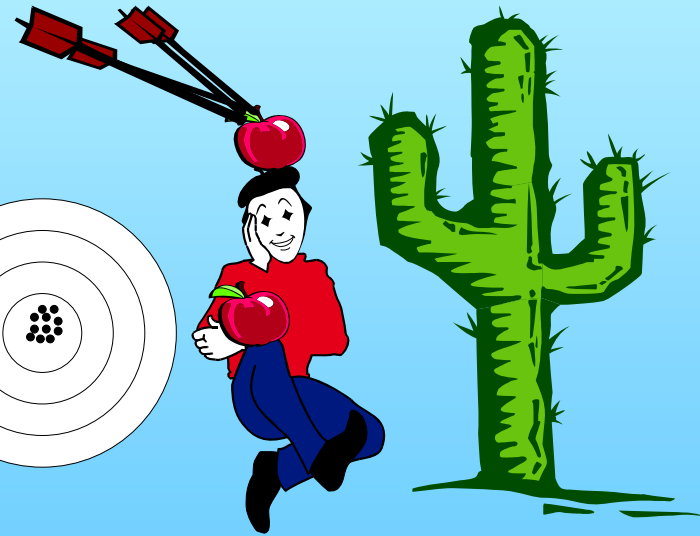


Good repeatability does
not necessary means good
accuracy
High precision, low accuracy









Good accuracy requires
good repeatability
High precision, high accuracy

More Instruments errors

Stability:

Instability is most likely to occur in instruments involving electronic processing with high degree of amplification. A common cause of this is adverse environment factors such as temperature and vibration.

Example, a rise in temperature may cause a transistor to increase the flow of current which in turn makes it hotter and so the effect grows and the display in played reading DRIFTS. In extreme cases the displayed value may jumped output. This may be caused by a poor electrical connection affected by vibration.

More Instruments errors

Time lag error:

In any instrument system, it must take time for a change in the input to show up on the indicated output. This time may be very small or very large depending upon the system. This is known as the response time of the system. If the indicated output is incorrect because it has not yet responded to the change, then we have time lag error.

A good example of time lag error is an ordinary glass thermometer. If you plunge it into hot water, it will take some time before the mercury reaches the correct level. If you read the thermometer before it settled down, then you would have time lag error.

More Instruments errors

Reliability:

Most forms of equipment have a predicted life span. The more reliable it is, the less chance it has of going wrong during its expected life span. The reliability is hence a probability ranging from zero (it will definitely fail) to to 1.0 (it will definitely not fail).

Drift:

This occurs when the input to the system is constant but the output tends to change slowly. For example when switched on, the system may drift due to the temperature changes as it warms up.

Calibration procedure example…

- A pressure transmitter is to be calibrated. The input pressure range is 30 to 150 psi. The output from the transmitter is 4 to 20 mA. Develop a calibration procedure to calibrate the transmitter. Calculate the settings and output for each reading that needs to be taken. Refer to instruction manual on the detail calibration procedure.

CALIBRATION READING	INPUT PRESSURE PERCENTAGE	INPUT PRESSURE	CALCULATED OUTPUT CURRENT	ACTUAL OUTPUT CURRENT	PERCENT DIFFERENCE
1	0	30 psi	4.0 mA		
2	25	60 psi	8.0 mA		
3	50	90 psi	12.0 mA		
4	75	120 psi	16.0 mA		
5	100	150 psi	20.0 mA		

CALIBRATION READING	INPUT PRESSURE PERCENTAGE	INPUT PRESSURE	CALCULATED OUTPUT CURRENT	ACTUAL OUTPUT CURRENT	PERCENT DIFFERENCE
1	100	150 psi	20.0 mA		
2	75	120 psi	16.0 mA		
3	50	90 psi	12.0 mA		
4	25	60 psi	8.0 mA		
5	0	30 psi	4.0 mA		

Instrument signals

Signal transmission : Telemetry is the transmission and reception of information using various types of signals:

The standard instrumentation signals

- 3 to 15 psi
- 4 to 20 mA
- 1 to 5 Volts

Other signals (occasionally used)

- 3 to 27 psi
- 10 to 50 mA

Note: The standard electrical signal 4 to 20 mA:

- ✓ Not affected by line resistance
- ✓ Less susceptible to induced voltages and noise than voltage signals
- ✓ An open circuit is easily detected

Calibration to assure correct I/O signals …

Calibration Methods

- Calibration Methods
  - √ Sequential Test
  - √ Random Test

Calibration Methods

- *Sequential Test*
  - A sequential test applies a sequential variation in the input value over the desired input range.
  - This may be accomplished by increasing the input value (upscale direction) or by decreasing the input value (downscale direction) over the full input range.

Calibration Methods

- *Random Test*
  - A random test applies a **randomly selected sequence of value of a known input** over the intended calibration range
  - The random application of input tends to minimize the impact of interferences. It breaks up hysteresis effects and observation errors

Calibration Methods

- A random test provides an important diagnostic test for the delineation of the several measurement system performance characteristics such as:-
    - Linearity error
    - Sensitivity error
    - Zero Shift error
    - Repeatability error

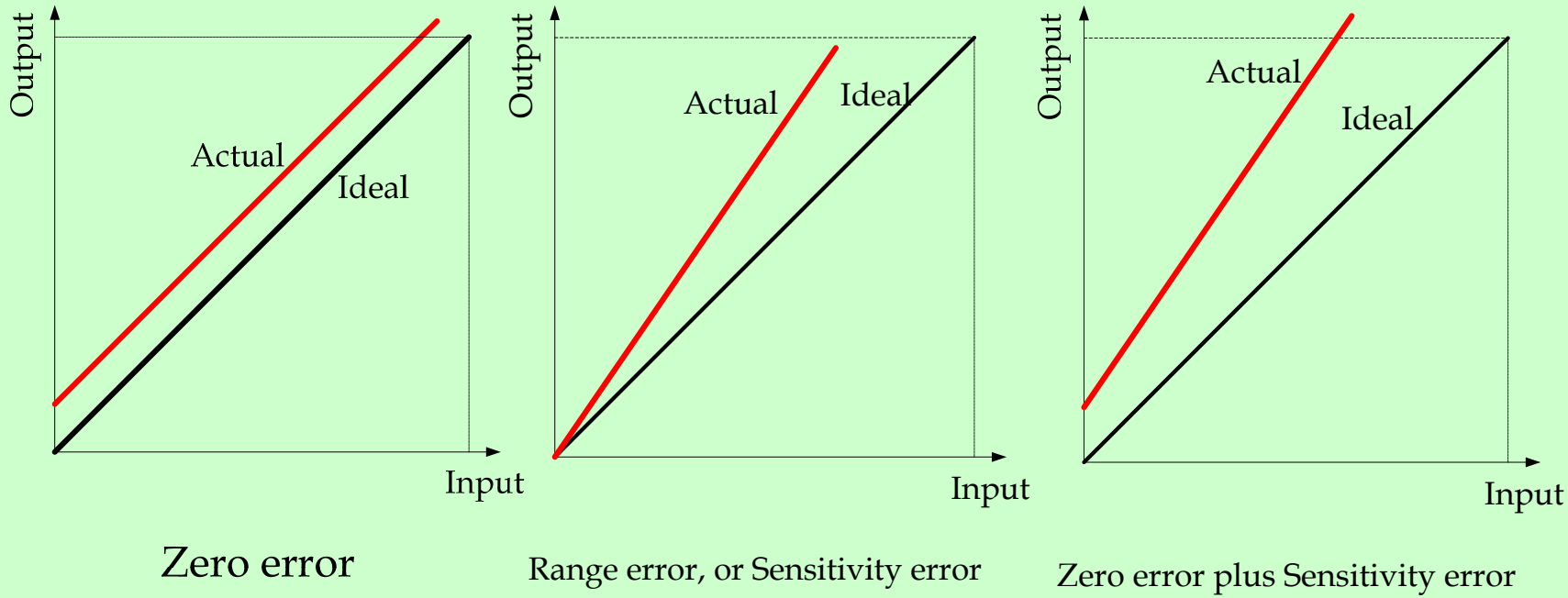
Calibration errors

Range and Zero Error:

After obtaining correct zero and range for the instrument, a calibration graph should be produced. This involves plotting the indicated reading against the correct reading from the standard gauge. This should be done in about ten steps with increasing signals and then with reducing signals. Several forms of error could show up. If the zero or range is still incorrect the error will appear as shown.

Calibration errors

Range and Zero Error:



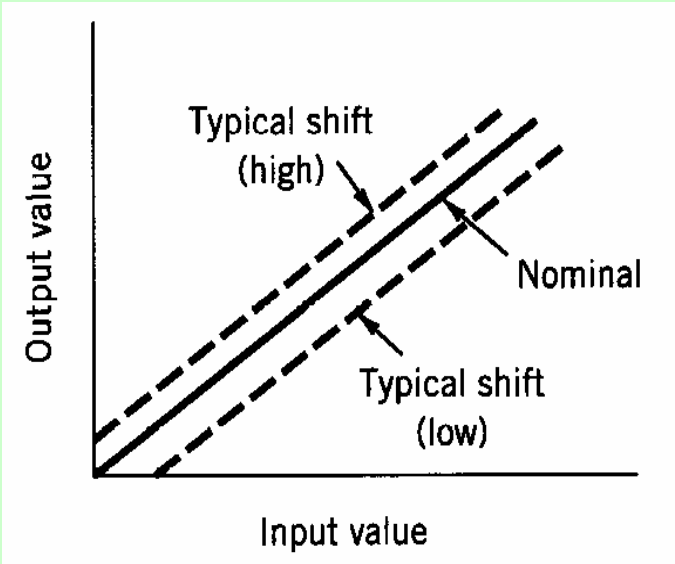
Zero error

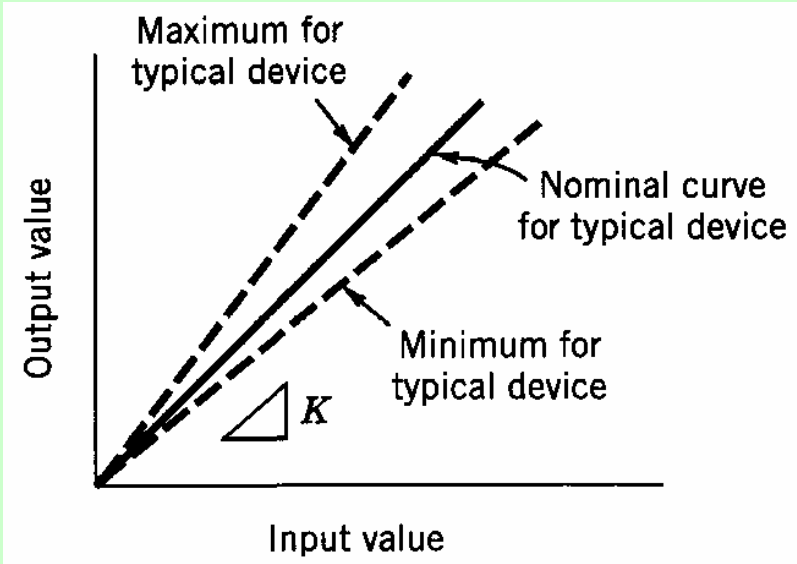
Range error, or Sensitivity error

Zero error plus Sensitivity error

Calibration errors

Range and Zero Errors:

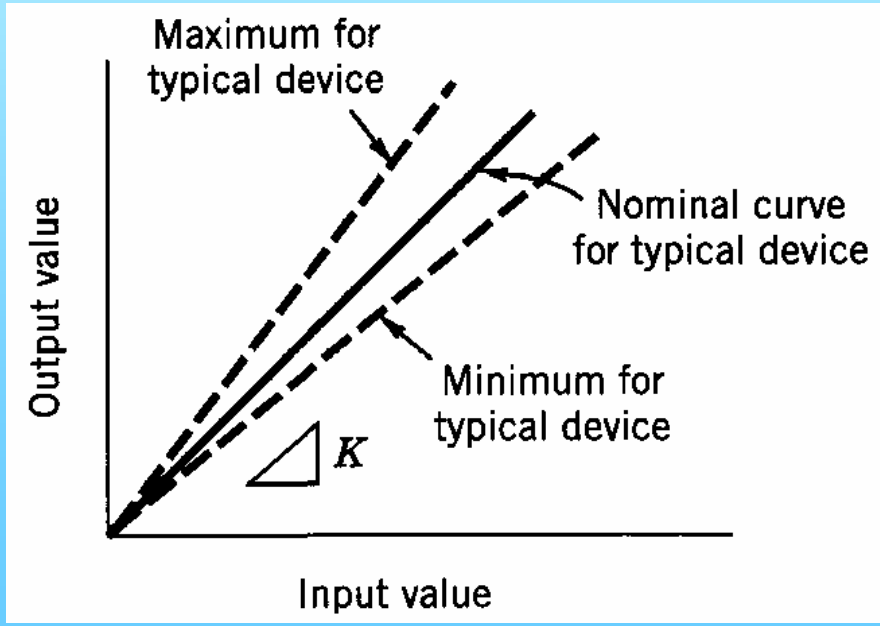




Zero error: The error in calibration at the lower end value.

Calibration errors

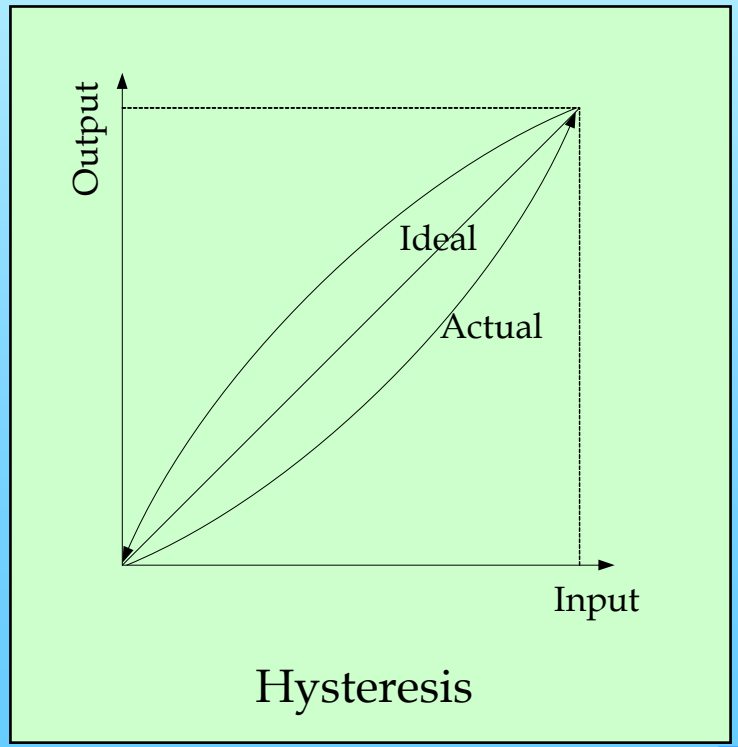
- *Static Sensitivity, K*
  - The slope of a static calibration curve yields the *static sensitivity(static gain)* of the measurement system.
  - *Sensitivity* is a ratio of a change of output magnitude to the change in input (after steady state).
  - Since calibration curves can be linear and nonlinear depending on the measurement system and on the variable being measured, *K* may or may not be constant over a range of input values



Calibration errors

HYSTERESIS

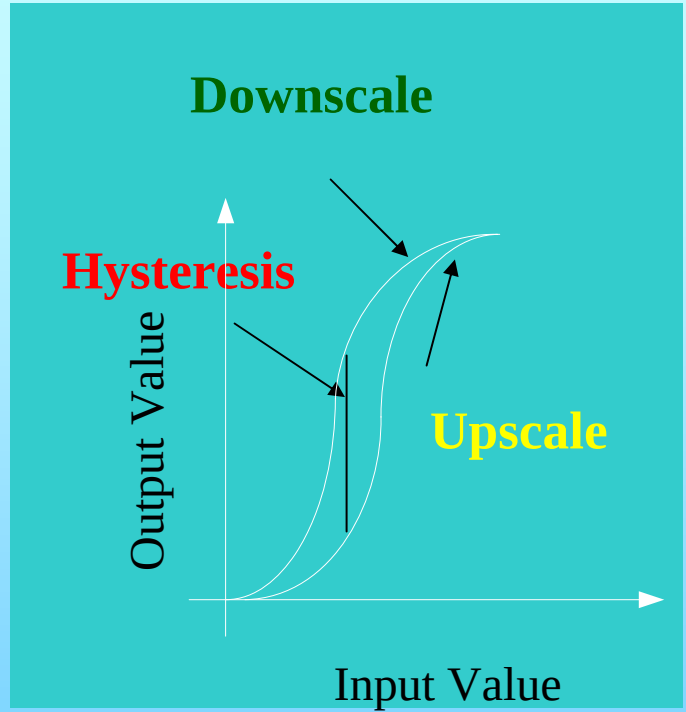
Hysteresis is produced when the displayed values are too small for increasing signals and too large for decreasing signals. This is commonly caused in mechanical instruments by loose gears and linkages and friction. It occurs widely with things involving magnetization and demagnetization



Hysteresis

Calibration errors

- *Hysteresis Error*
  - Refers to differences in the values found between going upscale and downscale in a sequential test.



$$e_h = (y)_{upscale} - (y)_{downscale}$$

Calibration errors

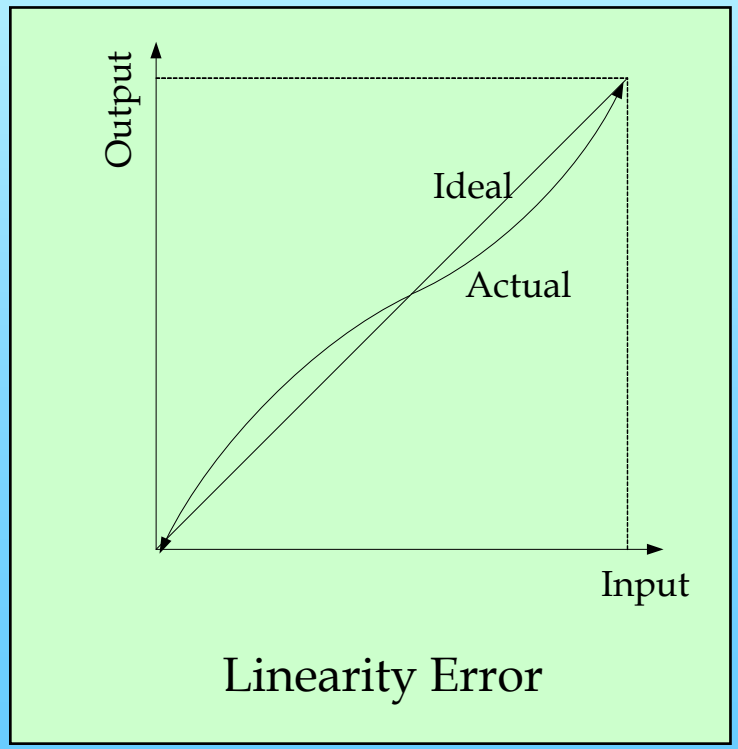
Hysteresis usually specified for a measurement in terms of the maximum hysteresis error found in the calibration, $e_{h\max}$, as a percentage of full-scale output range

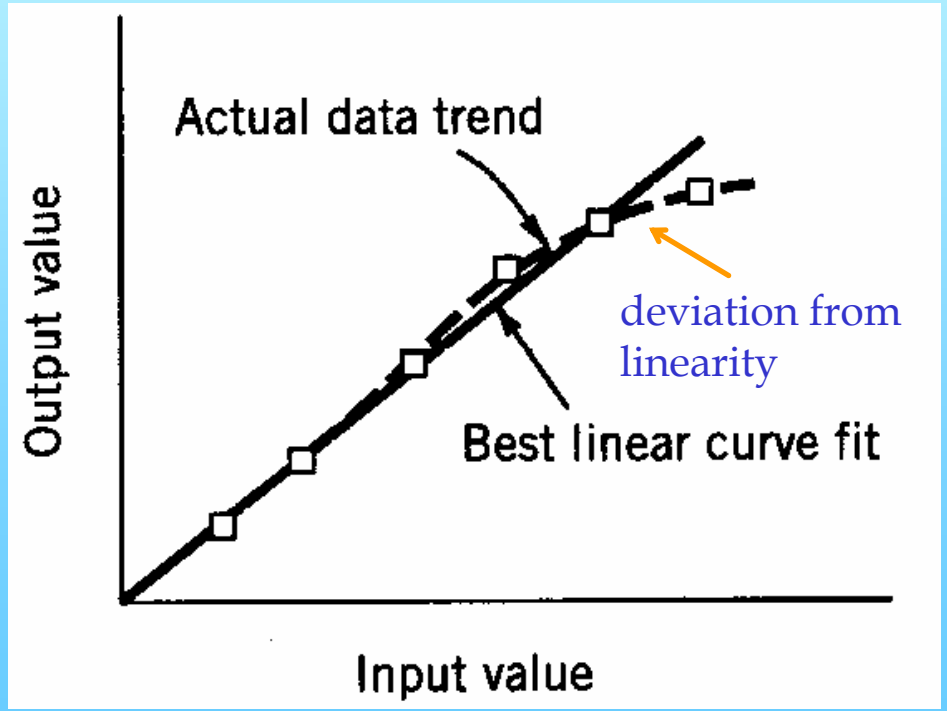
$$\%e_{h\max} = \frac{e_{h\max}}{r_o} \times 100$$

Calibration errors

NON LINEAR ERRORS

The calibration may be correct at the maximum values of the range but the graph joining them may not be a straight line (when it ought to be). This is a non linear error. The instrument may have some adjustments for this and it may be possible to make it correct at mid range as shown.





Example : Accuracy

Question: A sensor has a transfer function of 0.5 mV/^{o}C and an accuracy of ±1% FS. If the temperature is known to be 60°C, what can be said with absolute certainty about the output voltage?

Answer:

0.5 mV/^{o}C and an accuracy of ±1% FS means the transfer function could be 0.5±0.005mV/^{o}C or 0.495 to 0.505 mV/^{o}C.

At 60°C the output would be in the range (0.495 mV/^{o}C)(60°C) = 29.7mV to (0.505 mV/^{o}C(60°C) = 30.3 mV, or 30±0.3mV. That is, ± 1%.

Example : Zero drift, Sensitivity drift

Question: A sensor is calibrated in an environment at a temperature of 20°C and has the following deflection/load characteristic:

Load (kg)	0	1	2	3
Deflection (mm)	0	20	40	60

It is used in an environment at a temperature of 30°C and the following deflection/load characteristic is measured:

Load (kg)	0	1	2	3
Deflection (mm)	5	27	49	71

Example : Zero drift, Sensitivity drift

Determine the zero drift and sensitivity drift per ^{o}C change in ambient temperature.

Solution:

At 20°C, deflection/load characteristic is a straight line. Sensitivity = 20mm/kg.

At 30°C, deflection/load characteristic is still a straight line.
Sensitivity = 22 mm/kg

Zero drift (bias) = 5 mm (no load deflection)

Sensitivity drift = 2 mm/kg

Zero drift/^{o}C = 5/10=0.5 mm/^{o}C

Sensitivity drift/^{o}C = 2/10 = 0.2 (mm per kg)/^{o}C.

Example: Instrument Errors

The pressure transmitter specified in the table is chosen to measure a nominal pressure of 395 kPa. The ambient temperature is expected to vary between 36° C and 42° C during the test. What is the overall instrument error?Determine the accuracy of the transmitter if the indicated value is equal to 2.65 Vdc.

Operation	
Input Range	0 – 1000 kPa
Output Range	1 – 5 Vdc
Temperature Range	0 – 50° C
Performance	
Linearity error	0.10% FSO
Hysteresis error	0.10% FSO
Repeatability error	0.15% FSO
Thermal drift error	0.13% FSO
Thermal Sensitivity error	0.12% of reading

Instrument error = $\pm[(0.001)^2 + [(0.001)^2 + [(0.0015)^2 + [(0.0013)^2 + [(0.0012)^2]^{1/2} = \pm 0.27\%$

395kPa would give output of 2.58Vdc.

Accuracy = [(2.65-2.58)/2.58] x 100%

= ±0.27%

End of Lecture notes on Calibration